

YEAR 7 MATHEMATICS · CHAPTER 10

Dot plots

Lesson 3 of 8



In this lesson you will learn to:

- read information from a dot plot
- draw a dot plot from a list of data
- describe a dot plot using the mode, clusters and outliers

Name _____

Class _____

Teacher _____

Before you start

You will need

Pencil

Ruler

Syllabus: MA4-DAT-C-01: classifies and displays data using a variety of graphical representations

WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1** Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2** At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

Lesson map

Page	What is on it	Done
3	Learning intentions, key terms, notes and Level 1 teacher examples	☐
4	Level 1: YOU DO practice questions	☐
5	Level 2: WE DO teacher examples and YOU DO practice questions	☐
6	Level 3: WE DO teacher examples	☐
7	Level 3: YOU DO practice questions	☐
8	Level 3: YOU DO practice questions	☐
9	Extension questions and the tear-off exit ticket	☐
10	Lesson summary: a worked example for every level	☐

YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

- 1** Order: 7, 3, 9, 3, 5

- 3** 6 out of 30 as a simplified fraction

- 2** How many 4s: 4, 1, 4, 6, 4

- 4** Which value is far from the rest: 5, 6, 5, 7, 21



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.03

Dot plots

Date _____

Learning intentions

We are learning to:

- read information from a dot plot
- draw a dot plot from a list of data
- describe a dot plot using the mode, clusters and outliers

Success criteria

I can:

- ☐ count how many times a value occurs and find the **mode**
- ☐ draw a neat dot plot with an even scale and a label
- ☐ spot an **outlier** and describe where the data **clusters**

Key terms Write the definition as your teacher explains it.

Dot plot

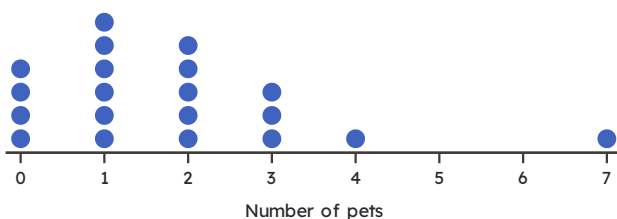
Outlier

Cluster

Notes

Level 1 · Getting started

Number of pets owned by students in 7B



WE DO

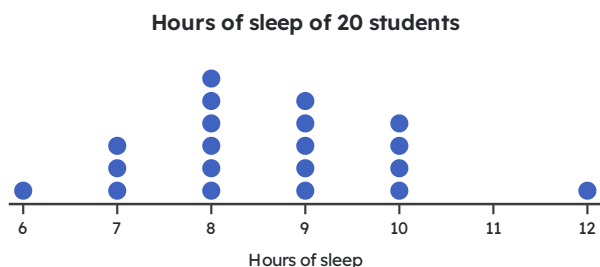
With your teacher: copy what they write on the board.

Example 1

How many students have 2 pets?
How many students were surveyed?

Example 2

What is the mode? Which value is an outlier?

YOU DO On your own: Set A. Use the sleep dot plot.

- 1 How many students were surveyed?
- 2 How many slept 8 hours?
- 3 How many slept 10 hours?
- 4 What is the mode?
- 5 What is the least sleep?
- 6 What is the most sleep?
- 7 What is the outlier?
- 8 How many slept 9 hours or more?
- 9 How many slept less than 8 hours?
- 10 How many more slept 8 hours than 7?

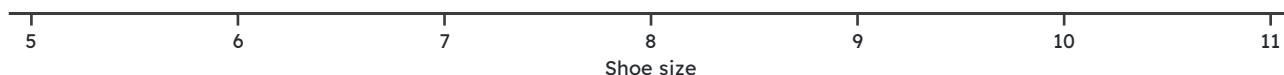
YOU DO On your own: Set B. Shoe sizes of 16 students.

Data: 6, 7, 7, 8, 8, 8, 9, 7, 8, 10, 9, 8, 7, 6, 8, 9

- 1 Complete the frequency table.

Size	6	7	8	9	10
Tally					
Frequency					

- 2 Draw a dot plot of the shoe sizes.

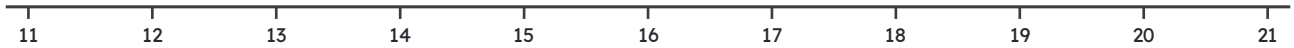


- 3 What is the mode?
- 4 How many wear size 9?
- 5 How many wear a size bigger than 8?
- 6 What is the smallest size?
- 7 How many wear size 7 or smaller?
- 8 Is there an outlier? (yes/no)
- 9 What fraction wear size 8?
- 10 Which size is least common?

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

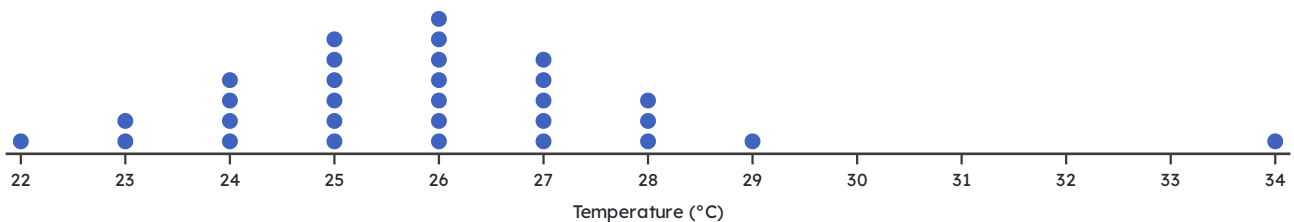
Example 3 Draw a dot plot for: 12, 14, 13, 12, 15, 12, 14, 20



Example 4 Describe the dot plot in Example 3. What is the mode? Where is the cluster? Is there an outlier?

YOU DO On your own: Set C. Use the temperature dot plot.

Daily maximum temperature, Kingscliff, November



1 What is the mode?

6 What percentage of days were 25 °C?

2 What is the outlier?

7 How many days were 24 or 25 °C?

3 How many days were 25 °C?

8 What fraction of days were 26 °C?

4 How many days were above 27 °C?

9 Where is the main cluster?

5 What fraction of days were below 24 °C?

10 Give a reason for the outlier.

**Level 3 · Challenge**

WE DO With your teacher: copy what they write on the board.

Test scores out of 10: 5, 6, 6, 7, 7, 7, 7, 8, 8, 8, 9, 9, 10, 2, 6, 7, 8, 9, 7, 8

Example 5 Draw a dot plot of the test scores. Choose your own scale and label it.



Write your own scale under the line, then label the axis.

Example 6 Describe the shape of the data. What does the dot plot tell the teacher about the class?

YOU DO

On your own: Set D. Text messages sent in one day by 25 students.

Data: 5, 7, 3, 6, 5, 12, 6, 4, 5, 8, 6, 7, 5, 30, 6, 5, 9, 7, 4, 6, 5, 8, 6, 7, 5

- 1 Draw a dot plot of the data. Choose a scale that fits every value, and label the axis.



Write your own scale under the line, then label the axis.

- 2 What is the mode?

- 4 How many students sent more than 7 texts?

- 3 Which value(s) are outliers?

- 5 What fraction sent 5 or fewer texts? Simplify.

YOU DO

On your own: Set D continued. Use your dot plot from the previous page.

- 6 What percentage sent exactly 6 texts?

- 7 Describe where the data clusters.

- 8 If the outlier is removed, does the mode change? Explain.

- 9 Why is a dot plot a good choice for this data?

- 10 Write one sentence that sums up what the dot plot shows.

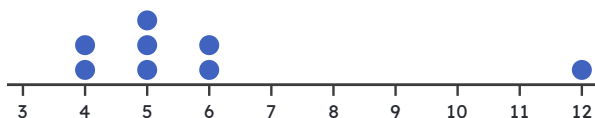
Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

Make up 8 values whose dot plot has a mode of 5, a cluster from 4 to 6, and an outlier at 12.

- Put the most dots at 5: 5, 5, 5
- Add values close by for the cluster: 4, 4, 6, 6
- Add one value far away for the outlier: 12



Answer: 4, 4, 5, 5, 5, 6, 6, 12

YOU DO

On your own: read the worked example, then try these.

- E1** Make up 10 values with a mode of 3 and one outlier. Draw the dot plot.

Write your own scale under the line, then label the axis.

- E2** Of 30 values, $\frac{1}{3}$ are 7s. There are twice as many 7s as 8s. How many 8s?

- E3** Why would a dot plot **not** suit the heights of 200 students?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

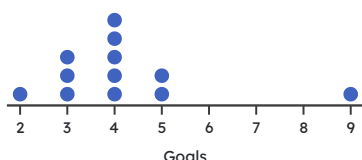
YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Goals in 12 netball games



Level 1

What is the mode?

Level 2

What fraction of the games had 3 goals or fewer? Simplify.

Level 3

Which value is the outlier? Where do the other values cluster?

Lesson summary

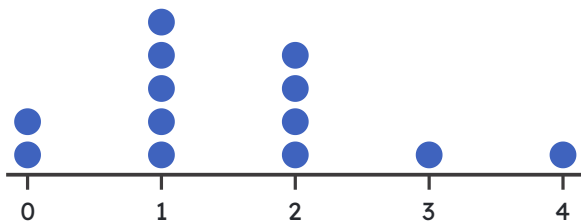
Missed the lesson? Start here.

- Drawing a dot plot**
1. Find the smallest and largest values.
 2. Draw a number line with an **even scale**.
 3. Put one dot per value above the line. Stack neatly.
 4. Add a **title** and label the axis.

Level 1

How many students have 2 siblings? What is the mode?

1. Count the dots above 2: there are 4
2. The tallest stack is above 1

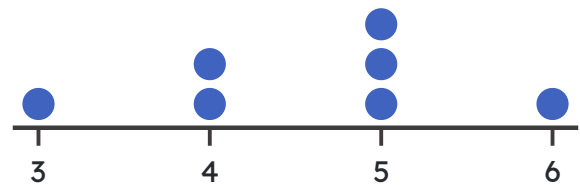


Answer: 4 students; mode 1

Level 2

Draw a dot plot for: 3, 5, 4, 5, 6, 5, 4

1. Smallest 3, largest 6: draw a line from 3 to 6
2. Put one dot above the line for each value

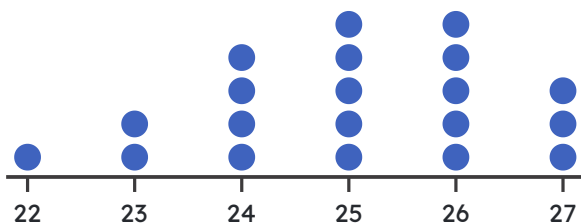


Answer: the dot plot below (mode 5)

Level 3

What fraction of these 20 days were warmer than 25 °C?

1. Count the dots above 26 and 27: $5 + 3 = 8$
2. Fraction = $8/20 = 2/5$

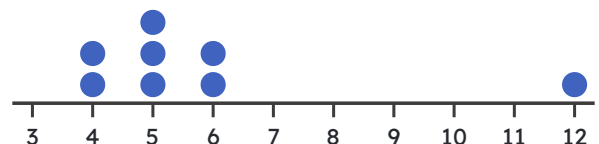


Answer: $2/5$ of the days (40%)

Extension

Describe the dot plot using a cluster and an outlier.

1. Most values bunch together from 4 to 6
2. 12 is far away from the rest



Answer: cluster from 4 to 6, outlier 12

This space is left blank on purpose. It is the back of your exit ticket.